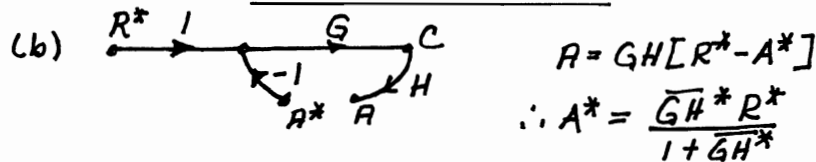


CHAPTER 5

5-1.(a) $C(s) = G(s) [R^*(s) - H^*(s) C^*(s)]$

$\therefore C(z) = \frac{G(z)}{1 + G(z)H(z)} R(z)$

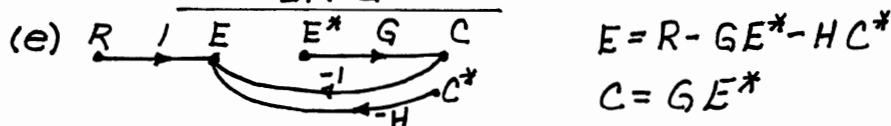
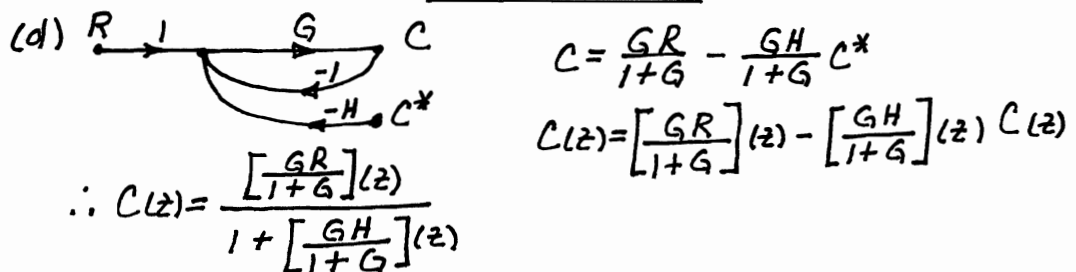


$\therefore C = G[R^* - A^*] = G\left[\frac{R^*}{1 + \overline{GH}^*}\right]$

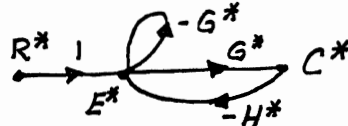
$\therefore C(z) = \frac{G(z)}{1 + \overline{GH}(z)} R(z)$

(c) $E = R - GHE^* \Rightarrow E^* = R^* - \overline{GH}^* E^* \Rightarrow E(z) = \frac{R(z)}{1 + \overline{GH}(z)}$

$C = GE^* \Rightarrow C(z) = \frac{G(z)}{1 + \overline{GH}(z)} R(z)$



$E^* = R^* - G^* E^* - H^* C^*$
 $C^* = G^* E^*$



$\therefore C(z) = \frac{G(z)}{1 + G(z) + G(z)H(z)} R(z)$

5-2.(a) $C = G_2 A^*, \therefore A = G_1 [R^* - \overline{G_2 H}^* A^*] \Rightarrow A(z) = \frac{G_1(z) R(z)}{1 + G_1(z) \overline{G_2 H}(z)}$

$\therefore C(z) = \frac{G_1(z) G_2(z)}{1 + G_1(z) \overline{G_2 H}(z)} R(z)$

$$5-2.(b) \quad C = G_1 G_2 [R^* - H^* C^*]$$

$$\therefore C(z) = \frac{\overline{G_1 G_2}(z)}{1 + \overline{G_1 G_2}(z) H(z)} R(z)$$

$$(C) \quad C = \frac{G_1 G_2 R}{1 + G_1 G_2 H} - \frac{G_2 H}{1 + G_1 G_2 H} C^*$$

$$\therefore C(z) = \frac{\left[\frac{G_1 G_2 R}{1 + G_1 G_2 H} \right](z)}{1 + \left[\frac{G_2 H}{1 + G_1 G_2 H} \right](z)}$$

$$(d) \quad C = \frac{GR}{1+GH_2} - \frac{GH_1}{1+GH_2} D^* C^*$$

$$\therefore C(z) = \frac{\left[\frac{GR}{1+GH_2} \right](z)}{1 + D(z) \left[\frac{GH_1}{1+GH_2} \right](z)}$$

(e)  $G_{2,1} = \left[\frac{1 - \epsilon - T_5}{5} \right] G_2$

$$A = G_1 R - G_1 G_2 H D^* A^*$$

$$\therefore A^* = \frac{\overline{G_1 R^*}}{1 + \overline{G_1 G_2 H^* D^*}} \quad \text{and} \quad C = G_2 D^* A^*$$

$$\therefore C(z) = \frac{G_2(z) D(z) \overline{G_1 R}(z)}{1 + \overline{G_1 G_2} H(z) D(z)}$$

5-3. (a) From Problem 5-1(b): $C(z) = \frac{G(z)}{1 + G_H(z)} R(z)$

(b) From Problem 5-1(c): $C(z) = \frac{G(z)}{1 + G(z)H(z)} R(z)$

(c) In each system, $E^* = R^* - \overline{GH}^* E^*$. Hence both $C_1(s)$ and $C_2(s)$ are the same for each system.

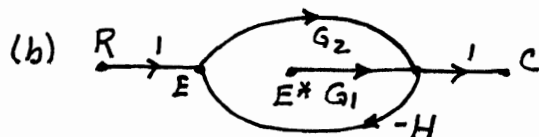
5-4.(a)

$$E = R - G, H E^*$$

$$\therefore E^* = \frac{R^*}{1 + GH^*}$$

$$5-4. (a) \quad C = G_2 R - G_1 G_2 H E^* + G_1 E^*$$

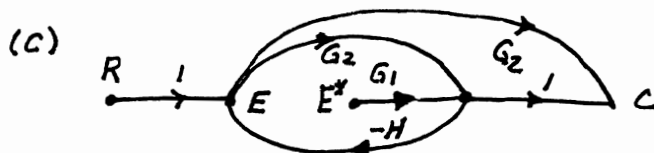
$$\therefore C(z) = \overline{G_2} R(z) + \frac{G_1(z) - \overline{G_1 G_2 H}(z)}{1 + \overline{G_1 H}(z)} R(z)$$



$$E = \frac{R}{1 + G_2 H} - \frac{G_1 H}{1 + G_2 H} E^* \Rightarrow E^* = \frac{\left[\frac{R}{1 + G_2 H} \right]^*}{1 + \left[\frac{G_1 H}{1 + G_2 H} \right]^*}$$

$$C = \frac{G_2 R}{1 + G_2 H} + \frac{G_1}{1 + G_2 H} E^*$$

$$\therefore C(z) = \left[\frac{G_2 R}{1 + G_2 H} \right](z) + \frac{\left[\frac{G_1}{1 + G_2 H} \right](z)}{1 + \left[\frac{G_1 H}{1 + G_2 H} \right](z)} \left[\frac{R}{1 + G_2 H} \right](z)$$



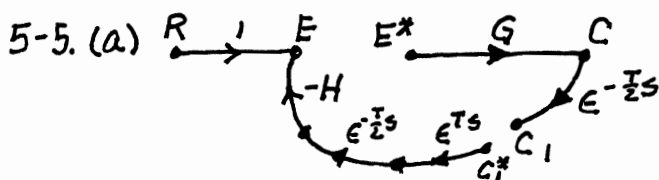
$$E = R - H[G_1 E^* + G_2 E] \Rightarrow E = \frac{R}{1 + G_2 H} - \frac{G_1 H}{1 + G_2 H} E^*$$

$$E^* = \frac{\left[\frac{R}{1 + G_2 H} \right]^*}{1 + \left[\frac{G_1 H}{1 + G_2 H} \right]^*}$$

$$C = G_2 E + G_1 E^* + G_2 E = 2G_2 E + G_1 E^*$$

$$= \frac{2G_2 R}{1 + G_2 H} + \frac{G_1 \left[\frac{R}{1 + G_2 H} \right]^*}{1 + \left[\frac{G_1 H}{1 + G_2 H} \right]^*}$$

$$\therefore C(z) = \left[\frac{2G_2 R}{1 + G_2 H} \right](z) + \frac{G_1(z) \left[\frac{R}{1 + G_2 H} \right](z)}{1 + \left[\frac{G_1 H}{1 + G_2 H} \right](z)}$$



$$E(z) = R(z) - z C_1(z) H(z, m) \Big|_{m=\frac{1}{z}}$$

$$C_1(z) = G(z, m) \Big|_{m=\frac{1}{z}} E(z)$$

$$\therefore E(z) = R(z) - z H(z, \frac{1}{z}) G(z, \frac{1}{z}) E(z)$$

$$C(z) = G(z) E(z) = \frac{G(z)}{1 + z G(z, \frac{1}{z}) H(z, \frac{1}{z})}$$

(b) Equivalent:

$$E = R \epsilon^{\frac{T}{2}s} - H C_1$$

$$C_1 = G E^* = G (R \epsilon^{\frac{T}{2}s})^* - G H C_1^*$$

$$\therefore C_1(z) = \frac{G(z) z [R \epsilon^{\frac{T}{2}s}]}{1 + \overline{G} H(z)} = \frac{G(z)}{1 + \overline{G} H(z)} z R(z, \frac{1}{z})$$

$$C_1(z) = C(\frac{T}{2}) + C(\frac{3T}{2}) z^{-1} + C(\frac{5T}{2}) z^{-2} + \dots$$

$$\therefore C(z) = C_1(z) z^{-\frac{1}{2}} = \frac{G(z)}{1 + \overline{G} H(z)} z^{\frac{1}{2}} R(z, \frac{1}{z})$$

$$= C(\frac{T}{2}) z^{-\frac{1}{2}} + C(\frac{3T}{2}) z^{-\frac{3}{2}} + \dots, \quad z^{\frac{1}{2}} \Big|_{z=\epsilon^{\frac{T}{2}s}} = \epsilon^{\frac{T}{2}s}$$

5-6.

$$X_1 = G_d \phi_P - G_d G_h G_a D^* X_1^* \Rightarrow X_1^* = \frac{\overline{G_d} \phi_P^*}{1 + D^* \overline{G_d} G_h G_a^*}$$

$$\therefore \phi_m(z) = \overline{G_h G_a}(z) D(z) X_1(z) = \frac{\overline{G_h G_a}(z) \overline{G_d} \phi_P(z) D(z)}{1 + \overline{G_d} G_h G_a(z) D(z)}$$

5-7.

$$X = -K G X^* + D^* K G_1 (R^* - C^*)$$

$$X(z) = \frac{D(z) K G_1(z)}{1 + K G_1(z)} [R(z) - C(z)]$$

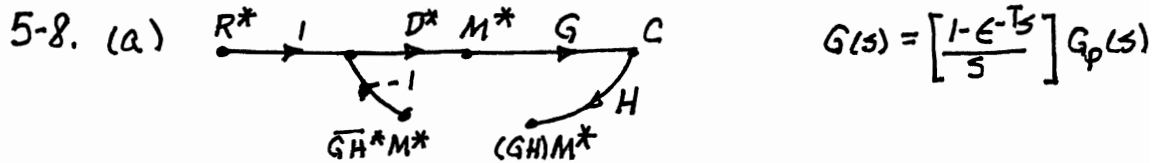
$$C = -K G_2 X^* + K D^* G_2 (R^* - C^*)$$

$$C(z) = -K G_2(z) \left[\frac{K D(z) G_1(z)}{1 + K G_1(z)} [R(z) - C(z)] + K D(z) G_2(z) [R(z) - C(z)] \right]$$

$$\therefore C(z) [1 + K G_1(z)] = -K G_2(z) D(z) K G_1(z) [R(z) - C(z)]$$

$$+ D(z) K G_2(z) [1 + K G_1(z)] [R(z) - C(z)]$$

$$5-7. \therefore C(z) = \frac{K D(z) G_2(z) R(z)}{1 + K G_1(z) + K D(z) G_2(z)}$$



$$M^* = D^*(R^* - \overline{G}H^*M^*) \Rightarrow M(z) = \frac{D(z)R(z)}{1 + D(z)\overline{G}H(z)}$$

$$C = GM^* \Rightarrow C(z) = \frac{D(z)G(z)}{1 + D(z)\overline{G}H(z)} R(z)$$

$$C(s) = G(s)M^*(s) = \frac{G(s)}{1 + D^*(s)\overline{G}H^*(s)} D^*(s)R^*(s)$$

(b) The signals out of the D/A's are identical.

5-9. (a) $G(s), H(s)$

(d) $H(s)$

(b) $G(s)$

(e) $G(s), H(s)$

(c) $G(s)$

5-10. (a) $G_1(s), G_2(s)$

(d) $H_1(s)$

(b) $G_1(s), H(s)$

(e) $G_2(s) = \left(\frac{1 - e^{-Ts}}{s} \right) G_{p2}(s)$

(c) $H_2(s)$

5-11. (a) $C(s) = G(s)R(s) - G(s)H(s)e^{-0.1Ts}D^*(s)C^*(s)$

$$\therefore C(z) = \overline{G}R(z) - \overline{G}H(z, m)|_{m=0.9} D(z)C(z)$$

$$\therefore C(z) = \frac{GR(z)}{1 + D(z)\overline{G}H(z, 0.9)}$$

(b) $C(z)$ same as in (a).

$$5-12.(a) \quad C = G D^* E^*$$

$$E = R - H_b G D^* E^* \Rightarrow E^* = \frac{R^*}{1 + D^* G^* H_b}$$

$$\therefore C(z) = \frac{D(z) G(z)}{1 + D(z) G(z) H_b} R(z)$$

$$(b) \quad C = -G_d R_d + G D^* E^*$$

$$E = -H_b G D^* E^* + H_b G_d R_d \Rightarrow E^* = \frac{H_b \overline{G_d R_d}^*}{1 + D^* G^* H_b}$$

$$C(z) = -\overline{G_d R_d}(z) + \frac{G(z) D(z) H_b \overline{G_d R_d}(z)}{1 + D(z) G(z) H_b} = \frac{-\overline{G_d R_d}(z)}{1 + D(z) G(z) H_b}$$

$$(c) \quad C(z) = \frac{D(z) G(z) R(z) - \overline{G_d R_d}(z)}{1 + D(z) G(z) H_b}$$

$$5-13.(a) \quad \pm 135^\circ \times 0.07 \text{ V/degree} = \pm 9.45 \text{ V}$$

$\therefore$ choose A/D with $\pm 10 \text{ V}$ input.

$$(b) \quad G(z) = \frac{z-1}{z} \mathcal{Z} \left[\frac{s}{s^2(0.5s+1)} \right]$$

$$M = D^* [\Theta_c - H_b K G M^*]$$

$$\therefore M^* = \frac{D^* \Theta_c^*}{1 + K D^* G^* H_b}$$

$$\Theta_d(z) = K G(z) M(z) = \frac{K D(z) G(z)}{1 + K D(z) G(z) H_b} \Theta_c(z)$$

$$(c) \text{ From Problem 4-15, } KG(z) = \frac{0.04488z + 0.04206}{z^2 - 1.8187z + 0.8187}$$

$$\begin{aligned} \therefore \frac{\Theta_d(z)}{\Theta_c(z)} &= \frac{0.04488z + 0.04206}{z^2 - 1.8187z + 0.8187 + 0.00314z + 0.00294} \\ &= \frac{0.04488z + 0.04206}{z^2 - 1.8156z + 0.8216} \end{aligned}$$

$$(d) \quad \begin{aligned} \text{num} &= [0 \ 0 \ 9.6]; \\ \text{den} &= [1 \ 2 \ 0]; \\ [Ac, Bc, C, D] &= \text{tf2ss}(\text{num}, \text{den}) \\ [A, B] &= \text{c2d}(Ac, Bc, .1) \\ [n, d] &= \text{ss2tf}(A, B, C, D) \\ \text{tnum} &= n \\ \text{tden} &= d + 0.07*n \end{aligned}$$

5-14. (a) $\theta_{\max}(t) = 360^\circ$, maximum sensor output = $(0.02)(360^\circ) = 7.2V$

$\therefore$ input for $360^\circ = 7.2V$

input for $0^\circ = 0V$

$\therefore$ need A/D range 0-7.2V, $\therefore$ choose 0-10V range.

5-14. (b) From (a), sensor output = $(0.02)(70^\circ) = \underline{1.4V}$

$\therefore$ input = 1.4V

(c) (a) maximum sensor output = $(0.02)(\pm 180^\circ) = \pm 3.6V$

$\therefore$ choose $\pm 5V$ range

(b) From (b), 1.4V

$$(d) G(z) = \mathcal{Z}\left[\frac{1-e^{-Ts}}{s} G_p(s)\right]$$

$$M^*(s) = D^*[R^* - KG^*H_b M^*] \Rightarrow M^* = \frac{D^*R^*}{1 + KD^*G^*H_b}$$

$$C = KG^*M^* \Rightarrow C(z) = \frac{KD(z)G(z)}{1 + KD(z)G(z)H_b} R(z)$$

$$(e) G(z) = \frac{z-1}{z} \mathcal{Z}\left[\frac{1}{Js^3}\right] = \frac{z-1}{z} \frac{(1)^2 z(z+1)}{0.1(2)(z-1)^3} = \frac{5(z+1)}{(z-1)^2}$$

$$\therefore \frac{KD(z)G(z)}{1 + KD(z)G(z)H_b} = \frac{2[5(z+1)]}{z^2 - 2z + 1 + 10(z+1)0.02}$$

$$\therefore \text{system transfer function} = \frac{10(z+1)}{z^2 - 1.8z + 1.2}$$

(f)

```

num=[0 0 20];
den=[1 0 0];
[Ac,Bc,C,D] = tf2ss(num,den)
[A,B] = c2d(Ac,Bc,1)
[n,d] = ss2tf(A,B,C,D)
tnum = n
tden = d + 0.02*n
    
```

5-15. (a) Sensor output = $0.4(30^\circ) = 12$

$\therefore 30^\circ \rightarrow r(k)=12$

$-30^\circ \rightarrow r(k)=-12$

$$5-15 (b) M^* = D^* (R^* - H_k^* K G^* M^*) ; \quad G(s) = \frac{1 - e^{-Ts}}{s} G_p(s)$$

$$\therefore M^* = \frac{D^* R^*}{1 + K D^* G^* H_k^*}$$

$$C(s) = K G(s) M^*(s) \Rightarrow C(z) = \frac{K D(z) G(z)}{1 + K D(z) G(z) H_k} R(z)$$

(c) From Problem 4-18,

$$G(z) = \frac{0.02268z + 0.02052}{z^2 - 1.7408z + 0.7408}$$

$$\begin{aligned} \therefore \frac{C(z)}{R(z)} &= \frac{0.4536z + 0.4104}{z^2 - 1.7408z + 0.7408 + 0.1814z + 0.1642} \\ &= \frac{0.4536z + 0.4104}{z^2 - 1.5594z + 0.9050} \end{aligned}$$

(d)

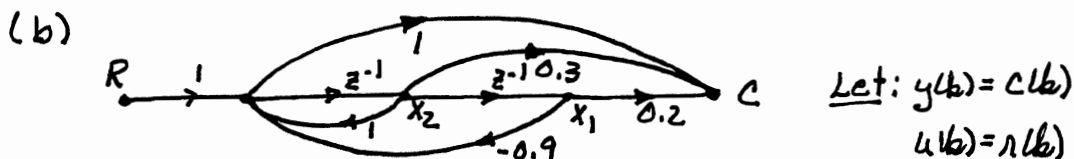
```

num=[0 0 400];
den=[1 6 0];
[Ac,Bc,C,D] = tf2ss(num,den)
[A,B] = c2d(Ac,Bc,.05)
[n,d] = ss2tf(A,B,C,D)
tnum = n
tden = d + 0.4*n
    
```

$$5-16. (a) \frac{C(z)}{R(z)} = \frac{1 + 0.3z^{-1} + 0.2z^{-2}}{1 - z^{-1} + 0.9z^{-2}}$$

$$(1 - z^{-1} + 0.9z^{-2})C(z) = (1 + 0.3z^{-1} + 0.2z^{-2})R(z)$$

$$\therefore c(k) = c(k-1) - 0.9c(k-2) + r(k) + 0.3r(k-1) + 0.2r(k-2)$$



$$\underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -0.9 & 1 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = [0.2 - 0.9 \quad 0.3 + 1] \underline{x}(k) + u(k) = [-0.7 \quad 1.3] \underline{x}(k) + u(k)$$

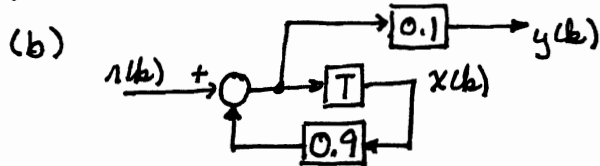
(c) $zI - A = \begin{bmatrix} z & -1 \\ 0.9 & z-1 \end{bmatrix}$, $|zI - A| = \Delta = z^2 - z + 0.9$
 $\therefore (zI - A)^{-1} = \frac{1}{\Delta} \begin{bmatrix} z-1 & 1 \\ -0.9 & z \end{bmatrix}$

$$5-16.(c) \frac{Y(z)}{U(z)} = C(zI-A)^{-1}B + D = \frac{1}{\Delta} [-0.7 \quad 1.3] \begin{bmatrix} z-1 & 1 \\ -0.9 & z \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} + 1$$

$$= \frac{1}{\Delta} [-0.7 \quad 1.3] \begin{bmatrix} 1 \\ z \end{bmatrix} + 1 = \frac{1.3z-0.7}{z^2-z+0.9} + 1 = \frac{z^2+0.3z+0.2}{z^2-z+0.9}$$

(d) $A = \begin{bmatrix} 0 & 1 \\ -0.9 & 1 \end{bmatrix}; B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}; C = [-0.7 \quad 1.3]; D = 1;$
 $[\text{num}, \text{den}] = \text{ss2tf}(A, B, C, D)$

5-17.(a) (a) $c(k+1) = 0.9c(k) + 0.1n(k+1)$

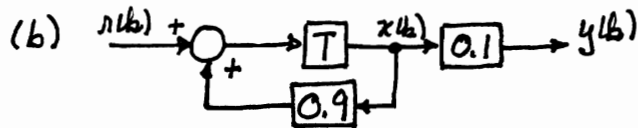


$$x(k+1) = 0.9x(k) + n(k)$$

$$y(k) = 0.09x(k) + 0.1n(k)$$

(c) $\frac{Y(z)}{R(z)} = C(zI-A)^{-1}B + D = \frac{0.09}{z-0.9} + 0.1 = \frac{0.1z}{z-0.9}$

(b) (a) $c(k+1) = 0.9c(k) + 0.1n(k)$

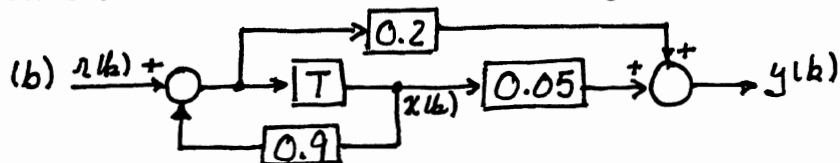


$$x(k+1) = 0.9x(k) + n(k)$$

$$y(k) = 0.1x(k)$$

(c) $\frac{Y(z)}{R(z)} = C(zI-A)^{-1}B = (0.1) \frac{1}{z-0.9} (1) = \frac{0.1}{z-0.9}$

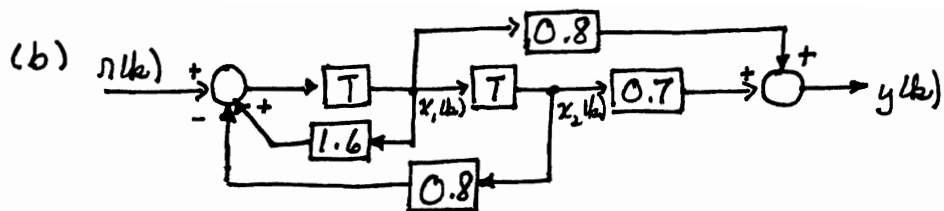
(c) (a) $c(k+1) = 0.9c(k) + 0.2n(k+1) - 0.05n(k)$



(c) $\frac{Y(z)}{R(z)} = C(zI-A)^{-1}B + D = \frac{0.13}{z-0.9} + 0.2 = \frac{0.2z-0.05}{z-0.9}$

(d) (a) $c(k+2) = 1.6c(k+1) - 0.8c(k) + 0.8n(k+1) + 0.7n(k)$

5-17.(d)



$$\underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -0.8 & 1.6 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} n(k)$$

$$y(k) = [0.7 \quad 0.8] \underline{x}(k)$$

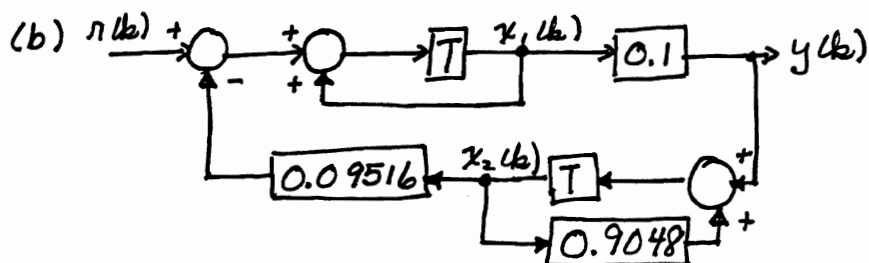
(c) $(zI - A) = \begin{bmatrix} z & -1 \\ 0.8 & z - 1.6 \end{bmatrix}$; $|zI - A| = \Delta = z^2 - 1.6z + 0.8$

$$\therefore \frac{Y(z)}{R(z)} = [0.7 \quad 0.8] \frac{1}{\Delta} \begin{bmatrix} z - 1.6 & 1 \\ -0.8 & z \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \frac{1}{\Delta} [0.7 \quad 0.8] \begin{bmatrix} 1 \\ z \end{bmatrix}$$

$$= \frac{0.8z + 0.7}{z^2 - 1.6z + 0.8}$$

5-18 (a) $G(z) = \frac{z-1}{z} \mathcal{Z}\left[\frac{1}{s^2}\right] = \frac{z-1}{z} \frac{0.1z}{(z-1)^2} = \frac{0.1}{z-1}$

$$H(z) = \frac{z-1}{z} \mathcal{Z}\left[\frac{1}{s^2 + s}\right] = \frac{z-1}{z} \frac{z(1 - e^{-0.1})}{(z-1)(z - e^{0.1})} = \frac{0.09516}{z - 0.9048}$$



(c) $\underline{x}(k+1) = \begin{bmatrix} 1 & -0.09516 \\ 0.1 & 0.9048 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} n(k)$

(d) $1 + G(z)H(z) = 0 = z^2 - 1.9048z + 0.9048 + 0.009516$

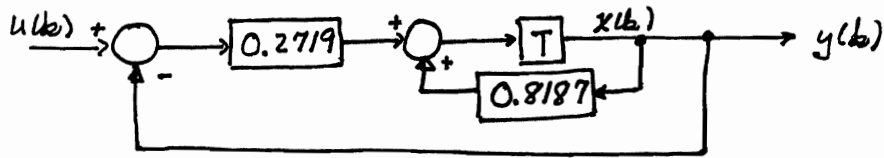
$$= z^2 - 1.9048z + 0.9143$$

(e) $|zI - A| = \begin{vmatrix} z-1 & 0.09516 \\ -0.1 & z-0.9048 \end{vmatrix} = z^2 - 1.9048z + 0.9048 + 0.009516$

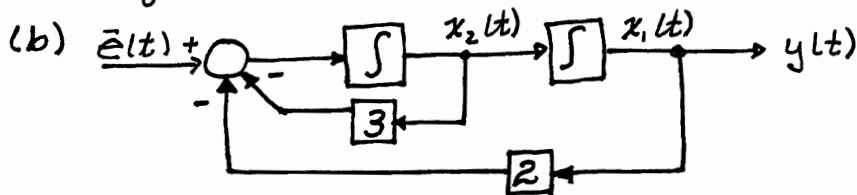
$$= \underline{z^2 - 1.9048z + 0.9143}$$

5-19. (a) $\frac{Y(s)}{M(s)} = G_P(s) = \frac{3}{s+2}$

$$G(z) = \frac{z-1}{z} \mathcal{Z}\left[\frac{3}{s(s+2)}\right] = \left(\frac{z-1}{z}\right) \frac{3}{z} \frac{z(1-e^{-2T})}{(z-1)(z-e^{-2T})} = \frac{0.2719}{z-0.8187}$$



$$\begin{aligned} \therefore x(k+1) &= [0.8187 - 0.2719]x(k) + 0.2719u(k) \\ &= 0.5468x(k) + 0.2719u(k) \\ y(k) &= x(k) \end{aligned}$$



$$\dot{x}(t) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 6 \end{bmatrix} \bar{e}(t)$$

$$y(t) = \begin{bmatrix} 1 & 0 \end{bmatrix} x(t)$$

$$\begin{aligned} (sI - A)^{-1} &= \begin{bmatrix} s & 1 \\ 2 & s+3 \end{bmatrix}^{-1} = \begin{bmatrix} \frac{s+3}{(s+1)(s+2)} & \frac{1}{(s+1)(s+2)} \\ \frac{-2}{(s+1)(s+2)} & \frac{s}{(s+1)(s+2)} \end{bmatrix} \\ &= \begin{bmatrix} \frac{2}{s+1} + \frac{-1}{s+2} & \frac{1}{s+1} + \frac{-1}{s+2} \\ \frac{-2}{s+1} + \frac{2}{s+2} & \frac{-1}{s+1} + \frac{2}{s+2} \end{bmatrix} \end{aligned}$$

$$\therefore \Phi_c(t) = \begin{bmatrix} 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \\ -2e^{-t} + 2e^{-2t} & -e^{-t} + 2e^{-2t} \end{bmatrix}$$

$$A = \Phi_c(0.1) = \begin{bmatrix} 0.9909 & 0.0861 \\ -0.1722 & 0.7326 \end{bmatrix}$$

$$\int_0^T \Phi_c(t) dt = \begin{bmatrix} -2e^{-t} + 0.5e^{-2t} & -e^{-t} + 0.5e^{-2t} \\ 2e^{-t} - e^{-2t} & e^{-t} - e^{-2t} \end{bmatrix}_0^{0.1}$$

$$B = \left[\int_0^T \Phi_c(t) dt \right] B_c = \begin{bmatrix} () & 0.00455 \\ () & 0.0861 \end{bmatrix} \begin{bmatrix} 0 \\ 6 \end{bmatrix} = \begin{bmatrix} 0.0273 \\ 0.5166 \end{bmatrix}$$

$$5-19. (b) \quad BC = \begin{bmatrix} 0.0273 \\ 0.5116 \end{bmatrix} \begin{bmatrix} 0 & 0 \end{bmatrix} = \begin{bmatrix} 0.0273 & 0 \\ 0.5116 & 0 \end{bmatrix}$$

$$\begin{aligned} \underline{x}(k+1) &= A \underline{x}(k) + B e(k) = A \underline{x}(k) + B [u(k) - y(k)] \\ &= [A - BC] \underline{x}(k) + B u(k) \end{aligned}$$

$$\therefore \underline{x}(k+1) = \begin{bmatrix} 0.9636 & 0.0861 \\ -0.6888 & 0.7326 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0.0273 \\ 0.5166 \end{bmatrix} u(k)$$

$$y(k) = \begin{bmatrix} 1 & 0 \end{bmatrix} \underline{x}(k)$$

$$5-20. \quad \underline{x}(k+1) = A \underline{x}(k) + B m(k)$$

$$y(k) = C \underline{x}(k) + D m(k)$$

$$m(k) = r(k) - y(k) = r(k) - C \underline{x}(k) - D m(k)$$

$$\therefore m(k) = [1/(1+D)] [r(k) - C \underline{x}(k)]$$

$$\therefore \underline{x}(k+1) = A \underline{x}(k) + B(1+D)^{-1} [r(k) - C \underline{x}(k)]$$

$$\therefore (1) \quad \underline{x}(k+1) = [A - (1+D)^{-1} BC] \underline{x}(k) + (1+D)^{-1} B r(k)$$

$$y(k) = C \underline{x}(k) + D(1+D)^{-1} [r(k) - C \underline{x}(k)]$$

$$\therefore y(k) = [C - D(1+D)^{-1} C] \underline{x}(k) + D(1+D)^{-1} r(k)$$

$$\therefore (2) \quad y(k) = (1+D)^{-1} C \underline{x}(k) + D(1+D)^{-1} r(k)$$

$$5-21. (a) \quad m(k) = r(k) - y(k) = r(k) - 0.2 x(k) - 0.5 m(k)$$

$$\therefore m(k) = 0.667 r(k) - 0.1333 x(k)$$

$$\therefore x(k+1) = 0.7 x(k) + 0.2 r(k) - 0.04 x(k)$$

$$y(k) = 0.2 x(k) + 0.3333 r(k) - 0.0667 x(k)$$

$$\therefore x(k+1) = 0.66 x(k) + 0.2 r(k)$$

$$y(k) = 0.1333 x(k) + 0.3333 r(k)$$

$$(b) \quad m(k) = r(k) - y(k) = r(k) - C \underline{x}(k)$$

$$\therefore \underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -0.9 & 1.3 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0.1 \\ 0.05 \end{bmatrix} [r(k) - 1.2 x_1(k) + 0.7 x_2(k)]$$

$$5-21.(b) \therefore \underline{x}(k+1) = \begin{bmatrix} 0.12 & 1.07 \\ -0.96 & 1.335 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0.1 \\ 0.05 \end{bmatrix} r(k)$$

$$y(k) = [1.2 \quad -0.7] \underline{x}(k)$$

$$(c) \quad m(k) = r(k) - C \underline{x}(k) = r(k) - x_1(k)$$

$$\therefore \underline{x}(k+1) = \begin{bmatrix} -0.5 & 0 & 0 \\ -1 & 0.9 & 1 \\ -3 & 0 & 0.9 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} r(k)$$

$$y(k) = [1 \quad 0 \quad 0] \underline{x}(k)$$

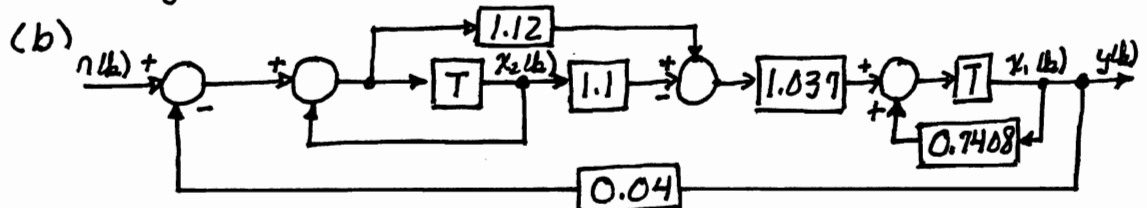
$$5-22. \quad D(z) = 1.1 + \frac{0.02z}{z-1} = \frac{1.12z - 1.1}{z-1}$$

$$(a) \text{ From Problem 4-21, } G(z) = \frac{1.037}{z - 0.7408}$$

$$\therefore \frac{D(z)G(z)}{1 + D(z)G(z)H} = \frac{1.1614z - 1.140}{z^2 - 1.6943z + 0.7001}$$

$$\therefore \underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -0.6952 & 1.6945 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} r(k)$$

$$y(k) = [1.1614 \quad -1.140] \underline{x}(k)$$



$$\underline{x}(k+1) = \begin{bmatrix} 0.6943 & 0.02074 \\ -0.04 & 1 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 1.1614 \\ 1 \end{bmatrix} r(k)$$

$$y(k) = [1 \quad 0] \underline{x}(k)$$

$$(c) \quad zI - A = \begin{bmatrix} z - 0.6943 & -0.02074 \\ 0.04 & z - 1 \end{bmatrix}, \Delta = |zI - A| = z^2 - 1.6943z + 0.6953$$

$$\frac{D(z)G(z)}{1 + D(z)G(z)H} = C(zI - A)^{-1}B = \frac{1}{\Delta} [1 \quad 0] \begin{bmatrix} z - 1 & 0.02074 \\ -0.04 & z - 0.6943 \end{bmatrix} \begin{bmatrix} 1.1614 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\Delta} [z - 1 \quad 0.02074] \begin{bmatrix} 1.1614 \\ 1 \end{bmatrix} = \frac{1.1614z - 1.140}{z^2 - 1.6943z + 0.6953}$$

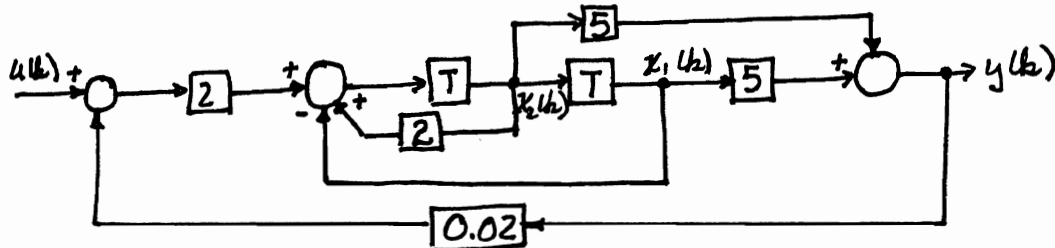
5.22. (a) The states of (a) are related to those of (b) by a similarity transformation: $\underline{x}_{(a)}(t) = P \underline{x}_{(b)}(t)$

5-23. (a) From Problem 5-14, $\frac{Y(z)}{U(z)} = \frac{10(z+1)}{z^2 - 1.8z + 1.2}$

$$\therefore \underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -1.2 & 1.8 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = [10 \quad 10] \underline{x}(k)$$

(b) From Problem 5-14: $G(z) = \frac{5(z+1)}{z^2 - 2z + 1}$



$$\underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -1.2 & 1.8 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0 \\ 2 \end{bmatrix} u(k)$$

$$y(k) = [5 \quad 5] \underline{x}(k)$$

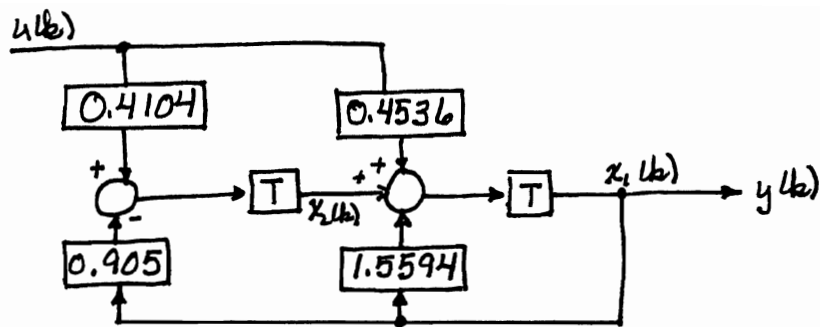
(c) $zI - A = \begin{bmatrix} z & -1 \\ 1.2 & z - 1.8 \end{bmatrix}$, $\Delta = |zI - A| = z^2 - 1.8z + 1.2$

$$\begin{aligned} \frac{Y(z)}{U(z)} &= C(zI - A)^{-1}B = [5 \quad 5] \frac{1}{\Delta} \begin{bmatrix} z - 1.8 & 1 \\ -1.2 & z \end{bmatrix} \begin{bmatrix} 0 \\ 2 \end{bmatrix} \\ &= [5 \quad 5] \frac{1}{\Delta} \begin{bmatrix} 2 \\ 2z \end{bmatrix} = \frac{10z + 10}{z^2 - 1.8z + 1.2} \end{aligned}$$

(d) $A = [0 \ 1; -1.2 \ 1.8]$; $B = [0; 2]$; $C = [5 \ 5]$; $D = 0$;
`[num, den] = ss2tf(A,B,C,D)`

5-24. (a) From Problem 5-15: $\frac{Y(z)}{U(z)} = \frac{0.4536z + 0.4104}{z^2 - 1.5594z + 0.9050}$

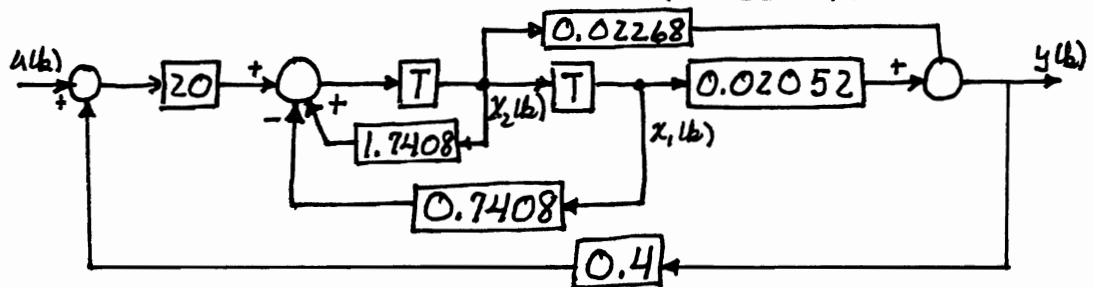
5-24.(a)



$$\underline{x}(k+1) = \begin{bmatrix} 1.5594 & 1 \\ -0.905 & 0 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0.4536 \\ 0.4104 \end{bmatrix} u(k)$$

$$y(k) = [1 \quad 0] \underline{x}(k)$$

(b) From Problem 5-15: $G(z) = \frac{0.02268z + 0.02052}{z^2 - 1.7408z + 0.7408}$



$$\underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -0.905 & 1.5594 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0 \\ 20 \end{bmatrix} u(k)$$

$$y(k) = [0.02052 \quad 0.02268] \underline{x}(k)$$

(c) $zI - A = \begin{bmatrix} z & -1 \\ 0.905 & z - 1.5594 \end{bmatrix}$, $\Delta = |zI - A| = z^2 - 1.5594z + 0.905$

$$\begin{aligned} \frac{Y(z)}{U(z)} &= C(zI - A)^{-1}B = [0.02052 \quad 0.02268] \frac{1}{\Delta} \begin{bmatrix} z - 1.5594 & 1 \\ -0.905 & z \end{bmatrix} \begin{bmatrix} 0 \\ 20 \end{bmatrix} \\ &= \frac{0.4536z + 0.4104}{z^2 - 1.5594z + 0.905} \end{aligned}$$

(d) $A = [0 \ 1; -0.905 \ 1.5594]$; $B = [0; 20]$;

$C = [0.02052 \ 0.02268]$; $D = 0$;

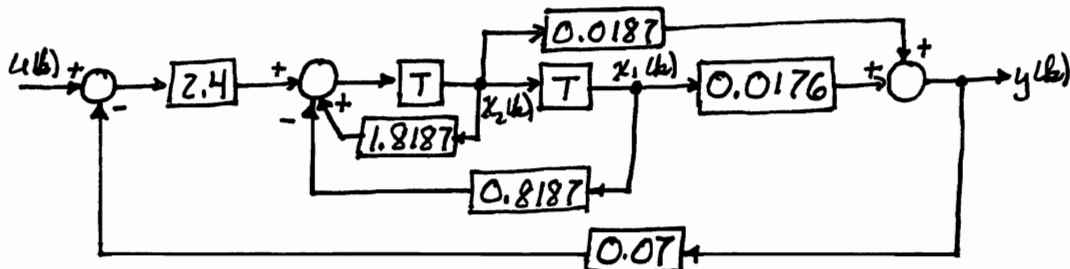
$[\text{num}, \text{den}] = \text{ss2tf}(A, B, C, D)$

5-25.(a) From Problem 5-13: $\frac{Y(z)}{U(z)} = \frac{0.04488z + 0.04206}{z^2 - 1.8156z + 0.8216}$

$$\underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -0.8216 & 1.8156 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(k)$$

$$y(k) = [0.04488 \ 0.04206] \underline{x}(k)$$

(b) From Problem 5-13, $G(z) = \frac{0.0187z + 0.0176}{z^2 - 1.8187z + 0.8187}$



$$\underline{x}(k+1) = \begin{bmatrix} 0 & 1 \\ -0.8217 & 1.8156 \end{bmatrix} \underline{x}(k) + \begin{bmatrix} 0 \\ 2.4 \end{bmatrix} u(k)$$

$$y(k) = [0.0176 \ 0.0187] \underline{x}(k)$$

(c) $zI - A = \begin{bmatrix} z & -1 \\ 0.8217 & z - 1.8156 \end{bmatrix}$, $\Delta = |zI - A| = z^2 - 1.8156z + 0.8217$

$$\frac{Y(z)}{U(z)} = C(zI - A)^{-1}B = \frac{1}{\Delta} C \begin{bmatrix} z - 1.8156 & 1 \\ 0.8217 & z \end{bmatrix} \begin{bmatrix} 0 \\ 2.4 \end{bmatrix}$$

$$= \frac{1}{\Delta} [0.0176 \ 0.0187] \begin{bmatrix} 2.4 \\ 2.4z \end{bmatrix} = \frac{0.04488z + 0.04224}{z^2 - 1.8156z + 0.8217}$$

5-26. (a) $\underline{w}(k+1) = A_1 \underline{w}(k) + B_1 m(k) + B_2 e(k)$

$$m(k) = C_1 \underline{w}(k) + D_1 e(k)$$

$$y(k) = C \underline{w}(k) + D_2 e(k)$$

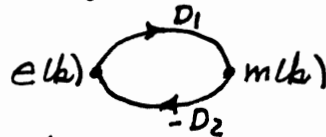
Solving,

$$\underline{w}(k+1) = \left[A_1 + \frac{D_2}{1 + D_1 D_2} (B_1 - D_2 B_2) (C_1 - D_1 C) - B_2 C \right] \underline{w}(k)$$

$$+ \left[B_2 + \frac{D_1}{1 + D_1 D_2} B_1 - \frac{D_1 D_2}{1 + D_1 D_2} B_2 \right] u(k)$$

$$y(k) = \left[C + \frac{D_2}{1 + D_1 D_2} (C_1 - D_1 C) \right] \underline{w}(k) + \frac{D_1 D_2}{1 + D_1 D_2} u(k)$$

5-26. (b) $m(k)$ is a direct function of $e(k)$, and $e(k)$ is a direct function of $m(k)$.



(c) The state equations do not exist.

(d) Delays cannot be ignored if this results in an algebraic loop with gain equal to -1 .